

Time: All in Your Mind

Clocks measure equal seconds. Human beings do not experience them equally. Research shows how mood, attention, memory and hope can reshape the felt movement of a life.

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Time appears to be one of the most objective features of existence. Clocks divide it into equal seconds, schedules organize it and calendars record its passage. Yet measured time is not the same as experienced time. An enjoyable afternoon may disappear almost unnoticed, while a period of waiting can seem interminable.

Psychologists distinguish between judging a duration and judging the passage of time. The first asks whether an interval lasted ten seconds or twenty. The second asks whether those seconds felt fast, slow, lively, empty or suspended. The difference is central to understanding why emotional states can change time without changing the clock.

Research on depression offers an especially revealing example. Findings do not support the simplistic claim that every depressed person loses the ability to estimate time. They suggest something subtler: depression can alter the felt direction of time, the emotional weight of the past and the accessibility of the future.

A clock can tell everyone in a room the same time. It cannot tell us whether that time feels alive with possibility.

Two kinds of time

Laboratory studies typically examine interval timing through tasks such as estimation, production, reproduction and discrimination. A participant might be asked to reproduce a tone's duration or decide which of two intervals was longer. These tasks test chronometric performance.

Passage-of-time judgments ask a different question: how quickly did time seem to move? The distinction helps resolve an apparent contradiction in the literature. People with depression often report that time drags, even when their performance on objective timing tasks remains broadly similar to that of non-depressed participants.

A meta-analysis by Thönes and Oberfeld combined 16 studies involving 433 participants with depression and 485 controls. Depression had a medium-sized effect on the subjective flow of time: participants with depression tended to experience time as passing more slowly. Yet the analysis found no significant overall effect on the four principal laboratory timing tasks (Thönes & Oberfeld, 2015).

16

Studies combined in the time-perception meta-analysis

918

Participants across depressed and control groups

$g = 0.66$

Effect on the subjective flow of time

The result suggests that a person can know how long an interval lasted while suffering through how that interval felt. Subjective time is not merely inaccurate clock time. It is an appraisal shaped by attention, emotion, activity, expectation and meaning.

The study of lived time

A 2018 study led by David Vogel investigated these experiences in depth. The researchers recruited 25 patients receiving inpatient treatment at the University Hospital Cologne. Participants had been diagnosed with a severe depressive episode and screened for non-chronic major depressive disorder (Vogel et al., 2018).

Instead of restricting participants to numerical scales, the researchers used an open-ended Time Questionnaire. It asked about the passage of time and about the meanings of the past, present and future. Participants could write as much or as little as they considered necessary.

The responses produced 235 statements, which the researchers organized into 30 categories through qualitative content analysis. Independent coding produced strong agreement, with a Krippendorff's alpha of .93 (Vogel et al., 2018). The study therefore focused not only on how quickly time seemed to pass, but on how time was lived from inside a life.

WHAT 'LIVED TIME' MEANS

Lived time concerns whether life feels as if it is progressing, standing still, repeating itself or moving beyond one's control. It includes the relationship among memory, present action and anticipated possibility.

A present that does not lead anywhere

Fifteen of the 25 participants reported a general slowing of time. Nine described its passage as inexorable, endless, circular or at a standstill. Five said that time had lost meaning or could no longer be properly felt, while six reported that pleasant or social activities could improve or normalize its passage (Vogel et al., 2018).

The present itself was unstable. Ten participants described it as extended, while another ten experienced it as a point rather than a duration. Nine connected it with negative emotions, and five described it as meaningless. Seven believed they could influence the present; seven believed it was outside their control (Vogel et al., 2018).

These apparently contradictory answers are informative. Depression does not produce one uniform clock. It can make the present feel both endless and strangely empty, or so compressed that it no longer functions as a bridge between past and future.

Ordinarily, the present receives some of its meaning from what it can change. People work now because the work may produce a later result. When the future no longer feels reachable, present action loses part of its direction. Chronological time continues, but psychological movement weakens.

“The problem may not be that the present is long. It may be that the present no longer seems to lead anywhere.”

Attention changes the clock we feel

A study by Sévigny, Everett and Grondin compared 15 participants classified as depressed with 20 non-depressed participants. The depressed group made more omissions on a sustained-attention task. They also performed more poorly when discriminating relatively long intervals of 1,120 and 1,280 milliseconds, while differences did not appear for shorter pairs of intervals. Their production of one- and ten-second intervals was also more variable (Sévigny et al., 2003).

The authors argued that the attentional demands of processing longer intervals may be important. This offers one possible mechanism: timing is not performed by an isolated internal clock. It depends partly on the cognitive resources available to monitor and organize an interval.

Gil and Droit-Volet used a temporal bisection task in which participants classified signals lasting between 400 and 1,600 milliseconds as short or long. Higher depression scores were associated with judging the signals as shorter, although

temporal sensitivity was similar across groups. Measures of sadness helped explain part of the pattern (Gil & Droit-Volet, 2009).

These experimental results should not be treated as perfectly consistent with one another. They use different samples, tasks and definitions. Their value lies partly in showing that 'time perception' is not a single ability. Attention, interval length, mood and task design can produce different outcomes.

Daily life is more complicated than a laboratory interval

A 2024 experience-sampling study followed 105 participants in daily life. Five times per day, participants evaluated how time had passed during the previous 30 minutes. At 11 p.m., they evaluated the day as a whole (Liu et al., 2024).

For the shorter 30-minute judgments, alertness, time expectation, emotional valence and arousal all predicted how quickly time seemed to pass. For judgments of the entire day, time expectation and emotional arousal remained predictive, while the other factors were less stable. The overall judgment of a day was most closely related to both the average and the most recent within-day reports (Liu et al., 2024).

The study highlights two principles. First, subjective time depends on more than whether a mood is positive or negative. Alertness and expectations matter. Second, retrospective judgments are constructed from an accumulation of moments but may be disproportionately influenced by what happened recently.

WHY EXPECTATION MATTERS

Time can feel slow when it violates our expectation of change. Waiting becomes especially difficult when we monitor time closely but cannot act to produce the event we want.

The downward spiral of waiting

Rankin, Sweeny and Xu studied undergraduate students waiting for examination grades and law graduates waiting for bar-exam results. Across three studies, people who experienced time as moving slowly tended to report more worry, anxiety, negative emotion and difficulty coping. The evidence also tentatively supported the reverse path: greater distress predicted a slower experience of time (Rankin et al., 2019).

This suggests a possible feedback loop. Distress makes time feel slower; slow time leaves more room to monitor uncertainty; that monitoring may intensify distress. The researchers described the association as potentially bidirectional, but their design did not establish a simple universal causal cycle (Rankin et al., 2019).

The finding extends beyond clinical depression. It shows that the emotional shape of time changes during ordinary periods of uncertainty. What matters is not only how long people wait, but whether they possess information, control and meaningful activity while waiting.

When the past becomes less specific

Depression is also associated with changes in autobiographical memory. Overgeneral autobiographical memory occurs when a person asked to recall one specific event instead produces a category of repeated experiences or an extended period. For example, rather than remembering one particular disappointing conversation, the person may recall that conversations 'always go badly.'

A 2023 meta-analysis by Weiss-Cowie, Verhaeghen and Duarte synthesized 67 published and unpublished works. Compared with controls, people with depression showed lower autobiographical-memory specificity, with Hedges' $g = -0.73$, and greater categoricity, with $g = 0.77$. The pattern was strongest in current clinical major depression but remained detectable in subthreshold and remitted groups (Weiss-Cowie et al., 2023).

Overgeneral memory does not mean that all memories are false. It concerns the level of detail and the ability to retrieve a bounded episode. Specific memories can support problem-solving because they provide concrete information about what

happened, what was tried and what changed. A general conclusion such as 'nothing ever works' offers fewer details from which to construct a new response.

67 Works synthesized in the memory meta-analysis	$g = -0.73$ Reduced autobiographical specificity	$g = 0.77$ Increased categorical remembering
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A future that is harder to picture

Remembering the past and imagining the future rely on related constructive processes. To picture a possible future, the mind recombines elements of previous experience. If specific details are difficult to retrieve, positive future scenes may also become less vivid or plausible.

Hallford and colleagues compared 117 people experiencing a major depressive episode with 47 participants without one. Participants imagined personally relevant positive future events. Those in the depressed group generated events with less specificity, detail, vividness, mental imagery and first-person perspective. They also rated those events as less plausible and reported less anticipatory pleasure (Hallford et al., 2020).

Within the depressed group, lower detail and vividness, weaker mental imagery and lower personal significance each uniquely predicted reduced pleasure while anticipating the event. The finding does not prove that imagery deficits cause anhedonia, but it shows that hope is partly sensory and constructive: a future that cannot be pictured clearly may be difficult to desire (Hallford et al., 2020).

This helps distinguish having no future from having a future that feels psychologically unavailable. A person may understand that positive events are possible in the abstract while struggling to simulate one as concrete, personal and likely.

“Hope is not only a belief that something good could happen. It is also the ability to imagine that event as vivid, personal and reachable.”

The idea of a blocked future

In Vogel and colleagues' study, ten participants associated the future with fear, worry or hopelessness. Eight described it as diminished, blocked, meaningless or no longer part of subjective experience. Seven characterized it as uncertain, impossible to plan or beyond their control (Vogel et al., 2018).

A blocked future does not necessarily mean that a person believes time will literally end. It means that future possibilities no longer exert their usual emotional pull. Advice such as 'look forward' or 'make a plan' assumes that the future is visible as a field of meaningful action. That assumption may not hold during a depressive episode.

The findings were not absolute. Ten participants also associated the future with hope - the same number that connected it with fear, worry or hopelessness. Contradictory relationships with the future can coexist. Depression can narrow possibility without erasing every positive expectation (Vogel et al., 2018).

What the evidence can and cannot establish

These studies differ substantially. Some recruited patients with diagnosed major depression; others measured depressive symptoms in non-clinical samples. Some examined milliseconds in a laboratory; others analyzed written descriptions, memories or imagined futures. Their results should not be collapsed into a single claim.

The Vogel study was small, qualitative and restricted to patients with severe depression who were receiving heterogeneous treatments. Its method was not designed to establish causation or represent every form of depression

(Vogel et al., 2018). The Thönes and Oberfeld meta-analysis noted that medication and dopamine-related mechanisms had not been investigated consistently enough to evaluate fully (Thönes & Oberfeld, 2015).

The future-thinking study did not assess cognitive functioning, and the comparison group's history of previous depressive episodes was unknown (Hallford et al., 2020). The daily-life study involved a non-clinical sample and described general influences on passage-of-time judgments rather than depression specifically (Liu et al., 2024).

These limitations do not make the evidence useless. They define what can responsibly be concluded: mood is associated with meaningful changes in subjective time, autobiographical memory and future simulation, but no single experience is universal and no one task captures the entire phenomenon.

A CLINICAL CAUTION

Research findings describe group-level tendencies. They cannot diagnose an individual, explain every episode of depression or replace assessment by a qualified professional.

Why subjective time matters

Subjective time affects more than comfort. If the present feels endless and the future inaccessible, making plans can become more difficult. If the past is remembered mainly in broad negative categories, it may offer fewer concrete exceptions and fewer usable solutions. If positive futures lack detail and plausibility, effort toward them may feel unrewarding.

Behaviors that look from the outside like passivity or a lack of motivation may therefore occur within a different temporal world - one in which the future does not exert its usual pull and the present does not feel capable of changing it.

The research also reveals something more general about perception. Human beings do not merely exist in time; they interpret it. Emotion gives time a pace, attention gives it density, memory gives it a history and imagination gives it a horizon.

Measured time remains indispensable. It coordinates societies and allows experiments to compare behavior. But lived time explains why two people can inhabit the same hour differently. One may experience an interval as temporary and meaningful; another may experience it as closed, repetitive and without an exit.

Understanding this difference encourages a more accurate form of empathy. It asks us not merely to tell another person that the future exists, but to recognize that the capacity to feel that future may itself be impaired.

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